

DDL: June 2nd, 2022

10.4 (a) Calculate $\Delta_r H^\ominus$ for the reaction $\text{Zn(s)} + \text{CuSO}_4(\text{aq}) \rightarrow \text{ZnSO}_4(\text{aq}) + \text{Cu(s)}$ from the information in Table 2.6 in the *Data section*.

10.7 (a) Relate the ionic strengths of (a) KCl, (b) FeCl_3 , and (c) CuSO_4 solutions to their molalities, b .

10.8 (b) Calculate the ionic strength of a solution that is $0.040 \text{ mol kg}^{-1}$ in $\text{K}_3[\text{Fe}(\text{CN})_6](\text{aq})$, $0.030 \text{ mol kg}^{-1}$ in $\text{KCl}(\text{aq})$, and $0.050 \text{ mol kg}^{-1}$ in $\text{NaBr}(\text{aq})$.

10.12 (a) Estimate the mean ionic activity coefficient and activity of a solution that is $0.010 \text{ mol kg}^{-1}$ $\text{CaCl}_2(\text{aq})$ and $0.030 \text{ mol kg}^{-1}$ $\text{NaF}(\text{aq})$.

10.16 (b) Consider a hydrogen electrode in aqueous HCl solution at 25°C operating at 105 kPa. Calculate the change in the electrode potential when the molality of the acid is changed from 5.0 mmol kg^{-1} to 50 mol kg^{-1} . Activity coefficients are given in Table 10.5.

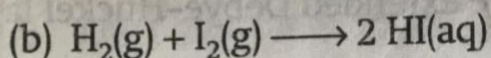
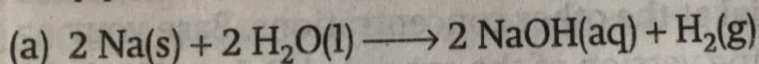
7.14(a) Write the cell reaction and electrode half-reactions and calculate the standard emf of each of the following cells:

(a) $\text{Zn} | \text{ZnSO}_4(\text{aq}) || \text{AgNO}_3(\text{aq}) | \text{Ag}$

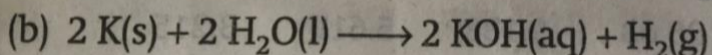
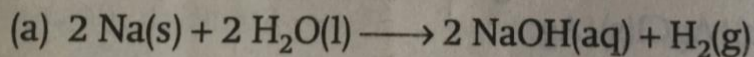
(b) $\text{Cd} | \text{CdCl}_2(\text{aq}) || \text{HNO}_3(\text{aq}) | \text{H}_2(\text{g}) | \text{Pt}$

(c) $\text{Pt} | \text{K}_3[\text{Fe}(\text{CN})_6](\text{aq}), \text{K}_4[\text{Fe}(\text{CN})_6](\text{aq}) || \text{CrCl}_3(\text{aq}) | \text{Cr}$

10.19 (b) Devise cells in which the following are the reactions:



10.22 (a) Calculate the standard Gibbs energies at 25°C of the following reactions from the standard potential data in Table 10.7:



10.24 (b) Consider the cell $\text{Pt} | \text{H}_2\text{(g, } p^\ominus) | \text{HCl(aq)} | \text{AgCl(s)} | \text{Ag}$, for which the cell reaction is $2 \text{AgCl(s)} + \text{H}_2\text{(g)} \rightarrow 2 \text{Ag(s)} + 2 \text{HCl(aq)}$. At 25°C and a molality of HCl of $0.010 \text{ mol kg}^{-1}$, $E = +0.4658 \text{ V}$. (a) Write the Nernst equation for the cell reaction, (b) Calculate $\Delta_r G$ for the cell reaction. (c) Assuming that the Debye-Hückel limiting law holds at this concentration, calculate $E^\ominus(\text{AgCl, Ag})$.

10.25 (b) Use the information in Tables 10.7 to calculate the standard emf of the cell $\text{Ag} | \text{AgNO}_3\text{(aq)} || \text{Fe(NO}_3)_2\text{(aq)} | \text{Fe}$ and the standard Gibbs energy and enthalpy of the cell reaction at 25°C. Estimate the value of $\Delta_r G^\ominus$ at 35°C.

10.26 (a) Calculate the equilibrium constants of the following reactions at 25°C from standard potential data:

